

Decode Labs

Cyber Security Internship

PROJECT 2

Basic Encryption & Decryption

Using the Caesar Cipher Algorithm

Submitted By

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Batch

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1. Introduction

This project demonstrates the implementation of a basic encryption and decryption system using the Caesar Cipher algorithm in Python. The purpose of the project is to understand the fundamentals of cryptography and how data can be protected by converting readable text into encrypted text and restoring it back through decryption.

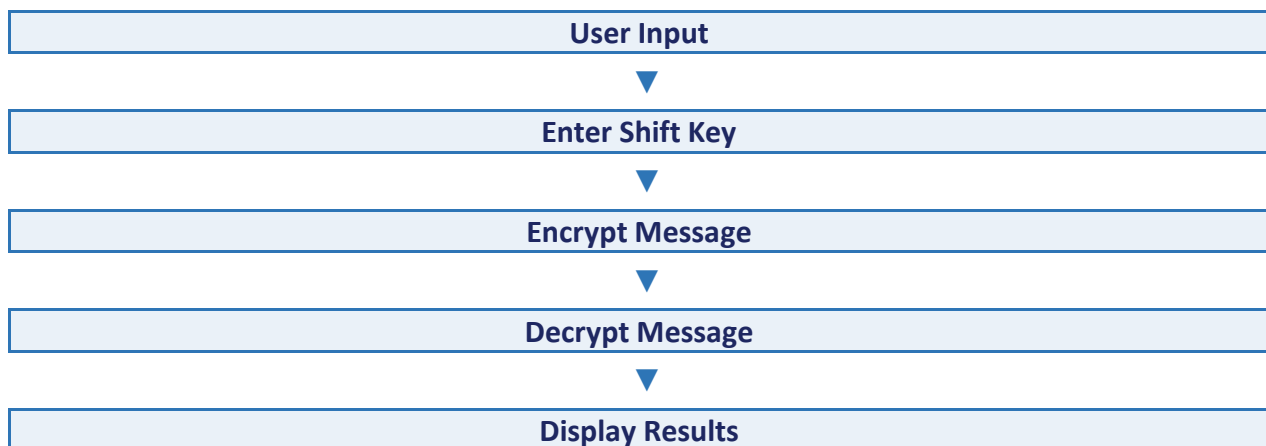
2. Objective

- Learn the basics of cryptography.
- Implement the Caesar Cipher using Python.
- Encrypt user input.
- Decrypt the encrypted message.
- Understand core encryption and decryption concepts.

3. Software Used

Software	Version	Purpose
Python	3.x	Programming Language
Visual Studio Code	Latest	Code Editor
Windows Terminal	—	Running the Program

4. Project Flow



5. Algorithm

- Read the message from the user.
- Read the shift key.
- Encrypt each alphabet character.
- Leave spaces and punctuation unchanged.
- Decrypt the encrypted message.
- Display the results.

6. Concepts Used

Encryption

Converts readable text (plaintext) into unreadable text (ciphertext) so that unauthorized users cannot understand it.

Decryption

Converts encrypted text back to the original readable message using the same key that was used to encrypt it.

Caesar Cipher

A classical substitution cipher in which each letter in the plaintext is shifted by a fixed number of positions in the alphabet.

`ord()`

A built-in Python function that converts a character into its corresponding ASCII integer value.

`chr()`

A built-in Python function that converts an ASCII integer value back into its corresponding character.

`% 26`

The modulo operation used to wrap letters around the alphabet, ensuring the shift stays within the 26 English letters.

7. Source Code

The following Python program implements the Caesar Cipher encryption and decryption logic.

```
Basic Encryption & Decryption.py M X
Task 2 > Basic Encryption & Decryption.py > encrypt
4 # =====
5
6 def encrypt(text, shift):
7     encrypted = ""
8
9     for char in text:
10         if char.isalpha():
11             if char.isupper():
12                 encrypted += chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
13             else:
14                 encrypted += chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
15             else:
16                 encrypted += char
17
18     return encrypted
19
20 def decrypt(text, shift):
21     decrypted = ""
22
23     for char in text:
24         if char.isalpha():
25             # Handle uppercase letters
26             if char.isupper():
27                 decrypted += chr((ord(char) - ord('A') - shift) % 26 + ord('A'))
28             # Handle lowercase letters
29             else:
30                 decrypted += chr((ord(char) - ord('a') - shift) % 26 + ord('a'))
31             else:
32                 decrypted += char
33
34     return decrypted
35
36 print("=" * 50)
37 print(" Caesar Cipher Encryption & Decryption")
38 print("=" * 50)
39
40 message = input("Enter your message: ")
41 shift = int(input("Enter shift key (1-25): "))
42
43 encrypted_text = encrypt(message, shift)
44 decrypted_text = decrypt(encrypted_text, shift)
45
46 938835ec*! (Rebasing) 0 0 Noor ul Ain (17 hours ago)
```

8. Output

The following screenshot shows the program running successfully in the terminal.

Sample Output

```
=====
      Caesar Cipher Encryption & Decryption
=====
Enter your message: DECODE
Enter shift key (1-25): 3

===== RESULT =====
Original Message : DECODE
Encrypted Message: GHFRGH
Decrypted Message: DECODE
=====
PS C:\Users\Microsoft\Desktop\Decode lab tasks> █
```

9. Conclusion

This project helped me understand the basic principles of cryptography by implementing the Caesar Cipher in Python. I learned how encryption and decryption work, how to use `ord()` and `chr()` for character conversion, and how modular arithmetic ($\% 26$) is used to wrap letters within the alphabet. Overall, this project strengthened my Python programming and problem-solving skills, and gave me a solid foundation for exploring more advanced cryptographic techniques.